

Vedanth Rai

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Objective

Enthusiastic final-year Computer Science undergraduate seeking opportunities to leverage my skills, bringing a strong foundational understanding of core computing principles and the ability to learn new technologies quickly, and eager to contribute.

Education

PES University, Bangalore
B.Tech in Computer Science

2023 – 2027

Experience

AI Intern – CellStrat

June 2025 – August 2025

- Worked with Python and SQL to build healthcare data pipelines, transforming raw patient data into structured information for downstream analysis.
- Developed real-time dashboards to surface key data and make analysis more accessible to users.
- Worked on optimizing an AI model inference pipeline, identifying processing bottlenecks and improving execution efficiency.

Projects

CodeOracle – Multi-Agent RAG Codebase Intelligence

- Built a multi-agent RAG system for GitHub repository understanding, coordinating 7 specialized agents for code retrieval, analysis, architecture, impact analysis, and verification.
- Implemented hybrid code retrieval using semantic embeddings, TF-IDF keyword search, and dependency-graph expansion, combined using Reciprocal Rank Fusion (RRF).
- Built code-aware static analysis using Python AST and dependency graphs, with LLM-based verification to ground generated answers against retrieved code.

Drishti – Trade Risk Intelligence

- Built an asynchronous multi-agent pipeline for agricultural trade-risk analysis using World Bank economic data, trade indicators, and GDELT news data.
- Implemented anomaly detection using StandardScaler and Isolation Forest, combined with Z-score analysis, volatility metrics, and linear-regression trend analysis.
- Developed a Streamlit dashboard for interactive risk visualization, with Docker deployment and a 30-test unit and integration test suite.

SkillBarter – Time-Banking & Skill Exchange Platform

- Built a full-stack Spring Boot MVC platform with JPA persistence, session management, skill matching, notifications, and role-based access control using Spring Security.
- Implemented a transactional credit-escrow workflow with state transitions for escrow, release, refund, and dispute resolution using Spring transactions.
- Applied Strategy, Observer, Builder, and Decorator patterns to create an extensible object-oriented architecture and separate core business responsibilities.

Technical Skills

- **Languages:** C, C++, Python
- **AI/ML:** PyTorch, Scikit-learn, LLM, Ollama
- **Databases:** MySQL, SQLite, ChromaDB
- **Core:** Data Structures & Algorithms, OOP, Operating Systems
- **Tools:** GitHub, VS Code, Jupyter, Google Colab, Streamlit

Achievements

- 1st Place – Nimbus 1000.
- Top 50 – Good Coder Competition.
- 3rd Place – Agentathon 2026.
- RBCCPS Internship – IISc Bangalore.